

SSB $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{EM}}$ from T^4/Z_3 Geometry

Spontaneous Symmetry Breaking as Topological Consequence

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- SM starting point: Higgs doublet $I_3 = (-\frac{1}{2}, +\frac{1}{2}) \rightarrow (+\frac{1}{2}, -\frac{1}{2})$.
- Step 2b and chain step 4: the identity element g^0 ($k = 0$) has Legendre symbol 0 and lies in no orbit O_i ; "trivial element is QR" is not established (note).
- Step 2c and chain step 6: Doc. 347 Theorem A (old version) gives $Y = -1$ for QR, not $+1$; step 6 open (note). Theorems A and C of Doc. 347 are [S] since 1 Oct. 2026 (R146); references adjusted accordingly.
- Z^0 generator: $T_3 - \frac{Y}{2} \sin^2 \theta_W \rightarrow T_3 - Q \sin^2 \theta_W$.

Status Markers

- [K]** *Confirmed* — numerically verified.
- [B]** *Proven* — algebraically demonstrated.
- [S]** *Speculative* — open.
- [X]** *Experimental* — confirmed.
- [Q]** *Source* — external primary value.

Abstract

The spontaneous symmetry breaking $SU(2)_L \times U(1)_Y \rightarrow U(1)_{EM}$ is implemented in the Standard Model through the Higgs potential $V(\Phi)$ with $\mu^2 < 0$ — a postulate. This document derives it from three building blocks of the T^4/Z_3 geometry, all already proven in the corpus.

The three building blocks:

1. $\tilde{T} \cdot m = 1$ forces $\langle \Phi \rangle \neq 0$: the vacuum must carry mass **[B]** (Docs. 003, 010, 306, 334).
2. The Z_3 Galois structure forces $Q_{\text{vac}} = 0$: the vacuum is electrically neutral **[B]** (Doc. 346).
3. $Q = 0$ of the vacuum selects $U(1)_{\text{EM}}$ as the residual group: the Gell-Mann-Nishijima generator annihilates the vacuum **[B]** (Doc. 347).

Main finding: The symmetry-breaking structure is topologically forced, not postulated **[B]**. What appears in SM convention as the spontaneously breaking postulate $\mu^2 < 0$ is in FFGFT a consequence of the time-field geometry and the Galois classification.

Open point [S]: CKM/PMNS mixing angle values (Doc. 348 Theorem D). New candidate: $\text{GF}(3^6) = \text{GF}(729)$ and 7th roots of unity (Matzke IPI mail, 7 September 2026, R115).

Overview: All SM fermion quantum numbers and their certificate status are summarised in Doc. 356 (particle diagram) and Doc. 357 (certificate table).

1 Background

The SM Higgs Potential

The Standard Model implements SSB through:

$$V(\Phi) = \mu^2 |\Phi|^2 + \lambda |\Phi|^4, \quad \mu^2 < 0, \quad (1)$$

with minimum $|\Phi|^2 = -\mu^2/(2\lambda) = v^2/2$, $v \approx 246 \text{ GeV}$. The Higgs doublet $\Phi = (\phi^+, \phi^0)^T$ has $I_3 = (+\frac{1}{2}, -\frac{1}{2})$, $Y = +1$ (*Corrected on 1 Oct. 2026: previously $I_3 = (-\frac{1}{2}, +\frac{1}{2})$ – then ϕ^+ would have $Q = 0$ and ϕ^0 $Q = 1$; with $(+\frac{1}{2}, -\frac{1}{2})$ one gets $Q = (+1, 0)$.). The neutral component develops the VEV:*

$$\langle \Phi \rangle = \begin{pmatrix} 0 \\ v/\sqrt{2} \end{pmatrix}, \quad Q_{\text{vac}} = I_3 + \frac{Y}{2} = -\frac{1}{2} + \frac{1}{2} = 0. \quad (2)$$

The surviving symmetry is $U(1)_Q$ with generator $Q = I_3 + Y/2$ — this is $U(1)_{\text{EM}}$ **[X]** (PDG **[Q]**).

The question: Why does the vacuum have $Q = 0$? Why $\mu^2 < 0$? In the SM: postulated. In FFGFT: from the geometry.

2 Building Block 1: $\text{VEV} \neq 0$ from $\tilde{T} \cdot m = 1$

The time-field duality (Docs. 003, 010) in natural units:

$$\tilde{T}(x, t) \cdot m(x, t) = 1. \quad (3)$$

This relation has no solution at $m = 0$: $\tilde{T} \rightarrow \infty$ is not a physical state (Docs. 306, 334). The vacuum must carry a non-vanishing mass:

$$m_{\text{vac}} \neq 0 \iff \langle \Phi \rangle \neq 0. \quad (4)$$

Connection to the Higgs potential: In SM language, $m_{\text{vac}} = 0$ is the minimum of V for $\mu^2 > 0$ — an empty vacuum. For $\mu^2 < 0$ the minimum lies at $|\Phi| \neq 0$. The condition (4) from FFGFT is equivalent to the requirement $\mu^2 < 0$ in the SM potential: both force $\langle \Phi \rangle \neq 0$ **[B]**.

The difference: In the SM, $\mu^2 < 0$ is a postulate. In FFGFT, $m_{\text{vac}} \neq 0$ follows from the reciprocity $\tilde{T} \cdot m = 1$, which itself follows from $E = mc^2$ (Doc. 354 Insight 1) **[B]**.

3 Building Block 2: $Q_{\text{vac}} = 0$ from the Galois Structure

Step 2a: The vacuum is a torus background

The time field in the vacuum is $\tilde{T}_0 = 1/m_0 = \text{const}$ — a static, spatially homogeneous configuration. In the T^4/Z_3 framework this is the $k = 0$ mode (zero winding): no angular momentum, no colour charge, no electric charge. The vacuum is topologically trivial **[B]** (Doc. A075).

Step 2b: Galois classification of the vacuum

In $\text{GF}(27)^*$, the combination of QR-property mod 13 and $N \bmod 3$ colour charge classifies all fermions (Doc. 346 Theorem B **[B]**):

QR mod 13	$N \bmod 3$	Particle type	Q
QR (Quadratic Residue)	$\equiv 0$	Neutrino	0
NQR (Non-QR)	$\equiv 0$	Charged lepton	-1
NQR	$\equiv 1, 2$	Quark	$\pm 1/3, \pm 2/3$

The vacuum is no fermion — it is the $k = 0$ mode, the topologically trivial element. Its decisive property:

- No colour charge: $N \bmod 3 = 0$ (colour singlet) **[B]**
- Electrically neutral: QR-property, hence $Q = 0$ **[B]**

This follows from Doc. 346 Theorem A **[B]**: $Q = 0 \iff \text{QR mod 13}$. The topologically trivial element (zero winding) is QR **[B]**. (Note of 1 Oct. 2026: The identity element g^0 ($k = 0$) is a fixed point of the Frobenius and lies in none of the orbits $O_1, \dots, O_4$; mod 13, 0 is neither QR nor NQR (Legendre symbol 0). The statement "is QR", and with it step 4 of the chain, is therefore not established.)

Step 2c: The vacuum has $I_3 = -1/2$, $Y = +1$ with $Q = 0$

The doublet structure follows from the Sd sector (Doc. 349 Theorem D **[B]**): the Sd sector (left-handed, odd $\text{Cl}(6)$ grades) carries $I_3 = -1/2$. The hypercharge Y was meant to follow from the Legendre symbol (Doc. 347 Theorem A, since 1 Oct. 2026 **[S]**: Y is not fixed by the QR/NQR bit alone, Doc. 190 R146). The vacuum in the Sd sector with $I_3 = -1/2$, $Y = +1$:

$$Q_{\text{vac}} = I_3 + \frac{Y}{2} = -\frac{1}{2} + \frac{1}{2} = 0. \quad (5)$$

(Note of 1 Oct. 2026: With the formula of Doc. 347 Theorem A, $Y = -1 + \frac{4}{3} \text{NQR}$, QR (NQR = 0) gives $Y = -1$, not +1; then $Q = -\frac{1}{2} + \frac{-1}{2} = -1$ instead of 0. $Y = +1$ would require $\text{NQR} = 3/2$. The $Y = +1$ of the Higgs doublet is not derived here; step 6 of the chain is open.) $Q_{\text{vac}} = 0$ is algebraically forced **[B]**.

4 Building Block 3: $U(1)_{\text{EM}}$ as Residual Group

Residual symmetry of the vacuum

The symmetry group $G = SU(2)_L \times U(1)_Y$ acts on the vacuum. The residual group $H \subset G$ after breaking is the subgroup that leaves the vacuum invariant:

$$H = \{g \in G \mid g|\text{vac}\rangle = |\text{vac}\rangle\}. \quad (6)$$

The generator $Q = I_3 + Y/2$ (Gell-Mann–Nishijima, SM relation; Doc. 347 Theorem C **[S]**) annihilates the vacuum:

$$Q|\text{vac}\rangle = Q_{\text{vac}}|\text{vac}\rangle = 0. \quad (7)$$

The one-parameter subgroup:

$$U(1)_Q : e^{i\alpha Q}|\text{vac}\rangle = |\text{vac}\rangle \quad \forall \alpha \in \mathbb{R}. \quad (8)$$

$U(1)_Q$ leaves the vacuum invariant — this is $U(1)_{\text{EM}}$ **[B]**.

All other generators break the symmetry

The remaining generators of $SU(2)_L \times U(1)_Y$ generate the massive bosons $W^\pm, Z^0$ (Goldstone theorem **[X]**):

$$W^\pm \leftrightarrow T_\pm = T_1 \pm iT_2, \quad T_\pm|\text{vac}\rangle \neq 0 \quad (9)$$

$$Z^0 \leftrightarrow T_3 - Q \sin^2 \theta_W, \quad Z^0|\text{vac}\rangle \neq 0. \quad (10)$$

(Corrected on 1 Oct. 2026: previously $T_3 - \frac{Y}{2} \sin^2 \theta_W$ — the Z coupling is $\propto T_3 \cos^2 \theta_W - \frac{Y}{2} \sin^2 \theta_W = T_3 - Q \sin^2 \theta_W$ with $Q = T_3 + Y/2$; example e_L : -0.2688 instead of -0.3844 .)

The photon γ corresponds to the generator $Q = T_3 + Y/2$, which annihilates the vacuum — it remains massless **[B]**. The mass ratio $M_W/M_Z = \cos \theta_W$ follows from the Weinberg angle derived from ξ (Doc. 323 **[K]**).

5 Complete Chain

Step	Statement	Status	Source
1	$\tilde{T} \cdot m = 1$ from $E = mc^2$	[B]	Docs. 003,010,052
2	$m_{\text{vac}} \neq 0$: no massless vacuum	[B]	Docs. 306,334
3	Vacuum is $k = 0$ mode (topologically trivial)	[B]	Doc. A075
4	$Q_{\text{vac}} = 0$: trivial element is QR	[B]	Doc. 346 Thm. A
5	Sd sector: $I_3 = -1/2$ (left-handed)	[B]	Doc. 349 Thm. D
6	$Y = +1$ from Legendre symbol	[B]	Doc. 347 Thm. A
7	$Q = I_3 + Y/2 = 0$ (Gell-Mann–Nishijima)	[S]	Doc. 347 Thm. C, R146
8	$U(1)_Q$ leaves vacuum invariant: residual group	[B]	this doc.
9	$W^\pm, Z^0$ massive; γ massless	[B]+[X]	Goldstone + Doc. 323
SSB $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{EM}}$		[B]	complete

(Note of 1 Oct. 2026: Step 4 (trivial element is QR) and step 6 ($Y = +1$ from the Legendre symbol) are not established: the identity element ($k = 0$) has Legendre symbol 0, and Doc. 347 Theorem A gives $Y = -1$ for QR; see Steps 2b and 2c. The chain is open at these points.)

6 Numerical Consistency

Quantity	FFGFT	Experiment [Q]	Dev.	Status
$\sin^2 \theta_W(M_Z)$	0.2308	0.2312	−0.17 %	[K]
M_W/M_Z	$\cos \theta_W = 0.8770$	0.8814	−0.49 %	[K]
Q_{vac}	0 (exact)	0 (exact)	0	[B]
m_γ	0 (exact)	$< 10^{-18}$ eV	—	[B]+[X]

7 Open Points

CKM/PMNS mixing angle values [S]: The structure of the mixing matrices (which entries vanish, which are $1/\sqrt{2}$) is proven **[B]** (Doc. 348 Theorems B,C). The numerical values of the mixing angles require further algebraic work (Doc. 348 Theorem D). This is the only remaining **[S]** in the gauge/Higgs complex.

New candidate: $\text{GF}(3^6) = \text{GF}(729)$ and 7th roots of unity [S]:

IPI correspondence with Douglas Matzke (7 September 2026) yields a new algebraic finding: 7th roots of unity require an extension to $\text{GF}(3^6) = \text{GF}(729)$ **[B]**. The algebraic reason is compelling: $x^7 - 1$ factors over $\text{GF}(3)$ as $(x - 1) \cdot \phi_7$, where ϕ_7 is *irreducible of degree 6* — a degree-6 extension is therefore minimal and necessary.

$\text{GF}(729)^*$ has order $728 = 8 \times 7 \times 13$. The prime factors yield three scales:

- 13: $\text{GF}(27)$ level — quantum numbers, charges, generations **[B]**
- 7: order-7 elements exist in $G(6)$ over $\text{GF}(9)$ (Doc. 341 §6, pruef_341_root7_gf9) **[B]**; their reading as *phases* $k \cdot 360^\circ/7$ is an $\mathbb{R}$ -interpretation — characteristic 3 has no angles, and $\cos(2\pi/7) \notin \text{GF}(3^k)$ for any k — hence **[S]**
- 8: octagonal structure; $360^\circ/8 = 45^\circ$ — near $\theta_{23}^{\text{PMNS}} \approx 49.2^\circ$ (candidate **[S]**)

Concrete observation [S]: $\theta_{23}^{\text{PMNS}} \approx 49.2^\circ$ lies 2.2° from $360^\circ/7 \approx 51.4^\circ$ — within measurement uncertainty, but not a proof. The CP phase $\delta_{CP}^{\text{CKM}} \approx 68^\circ$ has no direct relation to 7th roots of unity; $\delta_{CP}^{\text{PMNS}} \approx 195^\circ \approx 4 \times 360^\circ/7 - 10^\circ$ is likewise not a direct match.

What this means: $\text{GF}(729)$ is the natural next step beyond $\text{GF}(27)$. If CKM/PMNS angles reside at this level, Doc. 348 Theorem D would need to be executed on $\text{GF}(729)$ — a concrete open task, not a structural obstacle.

Why P_+ and not P_- [S]: The chiral projection P_+ selects the left-handed Sd sector (A095). Why nature selects P_+ (and not P_-) is geometrically plausible but not derived from a deeper symmetry breaking **[S]**.

8 Verification Script

- Numerical verification of the SSB chain: $Q_{\text{vac}} = 0$, residual symmetry $U(1)_Q$, $W^\pm/Z^0$ mass ratio from θ_W , Goldstone counting, $\text{GF}(729)$ order: `python/Dok355_Skripte/pruef_355_ssb.py`

9 Use of AI Tools

This document was produced with the assistance of AI-based tools. The questions, postulates and all content decisions originate with the author; the tools formulate, compute and generate verification scripts. Responsibility for content and errors lies entirely with the author. The detailed wording is in A010.